

CIS 371 Web Application Programming

Syllabus



Lecturer: **Dr. Yong Zhuang**

About this course

In this course, we dive deep into how websites and web apps are built and run. We'll look at the common methods and tools used in web development and study the different programming and markup languages that make websites work. We'll also cover important topics like how to connect websites to databases, making sure websites meet standard guidelines, and how to keep websites safe from security threats.

Course Objectives:

After completing this course, students should be able to:

- | Apply foundational knowledge of HTML to structure web content effectively.
- | Create stylesheets to control the visual appearance of web pages.
- | Craft modern front-end web applications using JavaScript and TypeScript.
- | Write server-side scripts to support the functionalities of a web application.
- | Utilize Ajax to fetch data from the server and dynamically update content without reloading the entire page.
- | Employ web APIs effectively to send and receive data between the client and server.
- | Analyze large-scale problems and construct comprehensive websites as solutions.

Contact Information

Instructor: Dr.Yong Zhuang

E-mail: yong.zhuang@gvsu.edu

Office: MAK D-2-234

Office Hours: MW, 11:00 am-12:00 pm, in-person (MAK D-2-234) and remote (Zoom)

Course Page: Blackboard & <https://gvsu-cis371.github.io>

Zoom: Meeting ID: 396 668 6420, Password: 587684

Section 01: **Class time:** Asynchronous

Room: Online

Final presentation: Week of April 27

Section 02: **Class time:** MWF 10:00 am - 10:50 am

Room: MAK B-1-124

Final presentation: (Monday) April 27, 10:00 am - 11:50 am

Course Logistics & Prerequisites

- No assigned textbook
 - most materials about web development is available online
- Check [Blackboard](#) on a regular basis for
 - announcements,
 - course material,
 - assignments
- Check [Course Website](#) regularly
- Zoom session (in case of snow days)
 - Login via <https://gvsu-edu.zoom.us>
 - Use the link and password posted on course website

Grading

Course Component	Overall Weight
Attendance Quizzes	5%
Quizzes	20%
Assignments	35%
Projects	40%
Total	100%

The instructor reserves the right to make minor adjustments to the point distribution.

Grading

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Assignments, Due Dates

-  **Due dates:** All assignments will be due at 11:59pm Michigan time on the due date.
-  **Late policy:** Each student is required to complete all learning activities by the due date deadline. No learning activities or assignments are accepted late. All assignments, graded discussions, quizzes, exams, etc., are submitted electronically. No assignments are accepted via email, printed, or any other method.

Attendance

- ★ Students are responsible for all material, announcements, and activities covered during class time. If you miss class, you must obtain lecture notes from a classmate.
- ★ This course uses brief in-class attendance quizzes to track student participation. These quizzes are not graded for correctness; they simply verify your attendance. You must submit the quiz during class time to receive credit.
- ★ Your attendance quiz grade contributes to 5% of your final course grade. Students who complete at least 80% of all attendance quizzes will receive full credit. If your attendance quiz submission rate is below 80%, you will receive zero for this portion.
- ★ There will be no make-up attendance quizzes unless prior arrangements are made or a documented excuse (e.g., illness, emergency) is provided.
- ★ Per University policy, "In case of excessive absences, the instructor may refuse to grant credit for the course." <https://www.gvsu.edu/catalog/navigation/academic-policies-and-regulations.htm#anchor-45>.

Academic Honesty

- 👉 Document Collaborations: Clearly note any collaborations on individual assignments.
- 👉 No Code Sharing: Direct electronic transfer of code between students is not allowed.
- 👉 Cite Internet Sources: If you use code from the internet, provide an active link and ensure it doesn't constitute the entire solution.
- 👉 Engage Online Respectfully: Participate in forums for discussion, not for sharing solutions or soliciting complete answers.
- 👉 Discuss Conceptually: Talk about problems using non-technical, conceptual language rather than sharing specific code.
- 👉 Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.

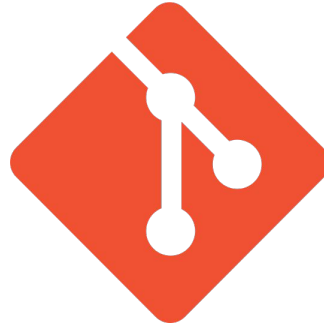
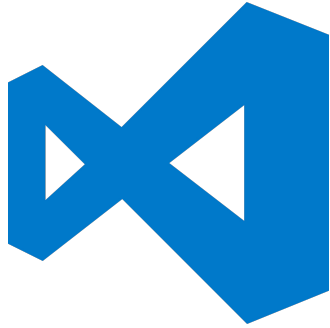
Prerequisites

- Formal Prerequisites: (CIS 162 or CIS 260) and (CIS 333 or CIS 353)
- Fluent in Java (or other object-oriented(OO) Languages)
 - You should be able to solve most problems at codingbat.com in a couple of minutes. If you struggle in solving these problems, then ***you are not ready*** to take this course
- Self Learner
 - Skilled in high-level programming **concepts**, and able to teach yourself the basics of other C-like languages (Java|Type)Script
- Good understanding of OO techniques: inheritance, methods, interface, ...

Expected Java Fluency

- Accessing object properties (without using a “getter” functions)
- Using loops on arrays of objects
- Writing own functions/methods
- Passing arguments into functions
- Function return value
 - Returning “result” from a function
 - Using a function “result”

Programming Tools



git



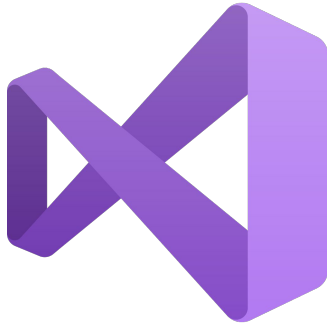
Debugger (in browser)



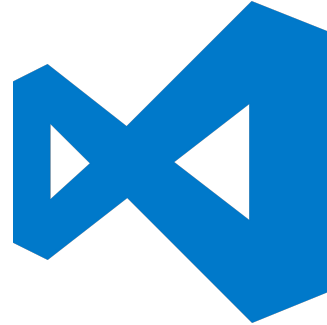
docker

Recommended

Programming Tools



Visual Studio



Visual Studio Code



Academic Resources

- | **The writing center:** The Fred Meijer Center for Writing, with locations at the Allendale and Pew/Downtown Grand Rapids campuses, is available to assist you with writing for any of your classes. For more information about these services and locations, please visit their website: <http://www.gvsu.edu/wc/>
- | **Speech lab:** The Grand Valley Speech Lab is a peer-to-peer communication center that helps students with all elements of oral presentations. For more information about this service, please visit their website: <https://www.gvsu.edu/speechlab/>

Academic Resources

- **Research consultants:** The Center for Scholarly and Creative Excellence (CSCE) promotes a culture of active, engaged, ethical scholarship. It supports innovative faculty and student research and collaborative partnerships in the broader community. For more information, please visit their website: <https://www.gvsu.edu/csce/>
- **Library:** GVSU's library offers a vast collection of online resources available for students. Visit their website for more details: <https://www.gvsu.edu/library/>
- **Disability support resources:** If any student in this class has special needs because of a disability, please contact Disability Support Resources at <http://www.gvsu.edu/dsr/> (DSR) at 616-331-2490.

Tentative Course Content

👉 January 19, Martin Luther King, Jr. Day Recess: No classes!

👉 March 8-14, Spring Break: No classes!

Week	Topics Covered
1	Syllabus, Introduction, Docker, HTML
2	Martin Luther King, Jr. Day (No Class) , Quiz 1, CSS 1
3	CSS 2, CSS Grid, Flexbox, Assignment 1
4	TypeScript 1, TypeScript 2, Quiz 3
5	Term Project, TypeScript 3, TypeScript 4
6	TypeScript 5, TypeScript 6, Quiz 4
7	HTTP, JavaScript Modules, Vue.js 1, Assignment 2
8	Vue.js 2, Quiz 5, Promise
9	Spring Break (No Class)
10	Pinia 1, Assignment 3, Pinia 2
11	Quiz 6, Vue Router, Vuetify 1
12	Assignment 4, Vuetify 2, Quiz 7
13	Firestore 1, Firestore 2, Firebase Authentication, Demo
14	Fetch, Axios, ExpressJS, Assignment 5
15	Assignment 5 Review, Project Lab, Project Lab
16	Final Project Presentation (April 27)

Course Website

***subject to change throughout the semester**

Warming Up

- Individual introduction
 - Name and what do you want to be called
 - Describe your background and experience in web development.
 - Specific topics you seek to learn from this class

About Me

- You can call me Dr. Zhuang (draw on), or Yong (you own)
- Education:
 - Ph.D., M.S. in Computer Science from the University of Massachusetts
- Experience/Interests:
 - My research interests include web application design, machine learning, and data mining.
- Personal Interests:
 - I enjoy spending time in city parks, hiking, and playing board games with friends and family.

